

Saberception: a gamified VR platform for perceptual research

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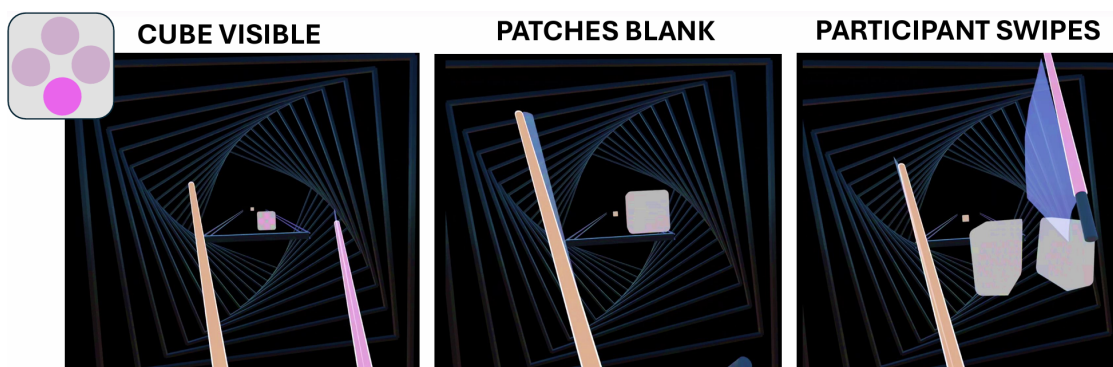


Figure 1: An example trial in our colour discrimination threshold task. Cubes spawn and move towards the participant at a rate of one cube/second. At a fixed distance, cubes make a jump towards torso height. The patches in which the odd-one-out indicates the correct slice direction appear for 500ms. Then the cube continues for another 500ms until sliced by the participant.

ABSTRACT

Saberception is a gamified, immersive VR platform for conducting perceptual experiments under more active and engaging conditions. Inspired by the VR game ‘BeatSaber’, participants slice through oncoming cubes in sync to music. Their performance as a function of stimuli intensity can be used to measure various perceptual thresholds. We demonstrate the utility of the method with a colour discrimination experiment, showing it is possible to rapidly and reliably measure perceptual thresholds during an active task in a fun and engaging manner.

1 INTRODUCTION

How do we measure the capabilities of human perception? Traditionally, perceptual research has relied on psychophysical methods in which restrained observers make repeated perceptual judgements about tightly controlled stimuli. These methods have resulted in a wealth of understanding about perceptual processing. There are however concerns about the generalizability of data collected under these controlled conditions to “real-world” perception. Psychophysical methods are often straining and repetitive, potentially leading to attentional lapses which can negatively affect the validity of estimated perceptual thresholds [6]. Further, in the real world our perceptual judgements are embedded within action and interaction. Recent work has demonstrated that perception may be altered under more ecological conditions such as moving stimuli [7] and to facilitate eye movements [8]. It is important therefore to explore perceptual processing during more active and engaging tasks.

To facilitate this we developed the ‘Saberception’ methodology. We developed a custom VR rhythm cube slicing game, inspired by the popular game ‘BeatSaber’. Participants slice through musically synchronized cubes that move towards them with virtual swords.

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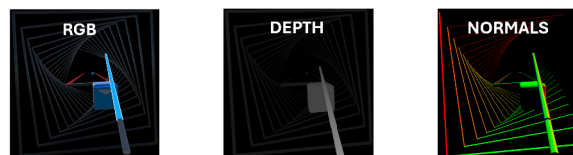


Figure 2: Example data available from the graphics engine. Gameplay recordings allow all frames viewed by participants to be regenerated and saved. The graphics engine provides access to potentially useful information such as depth maps and normals.

Stimuli on the cube faces indicate the direction participants should slice, and by varying the stimuli contrasts or intensities we can measure various perceptual properties such as contrast sensitivity and color discrimination. We posit that this method supports perceptual research during a more active task while increasing participant engagement and facilitating rapid and fun data collection.

2 THE SABERCEPTION PLATFORM

The core of the Saberception platform is a ‘BeatSaber’ clone developed with Unity and OpenXR. Researchers can design experiments by altering config files and graphical assets, allowing them to change cube motion properties, timings, appearance etc. The ‘BeatSaber’ game naturally lends itself to 4-AFC tasks (i.e. the correct slicing direction being determined by stimuli on the frontal face), however other options are available in the platform such as 2-AFC, categorization, and go/no-go tasks.

Currently there are two ways to define levels. They can be generated as a regular stream of cubes based on the timings and stimulus levels chosen by the researcher. Stimulus levels can either be fixed and randomized/counter-balanced throughout the level (e.g. method-of-constant stimuli), or stimulus levels can be dynamically changed via staircase procedures via network connection to PsychoPy/Psychtoolbox server. Alternatively, levels can be predefined in a JSON config file describing the spawn time, stimulus proper-

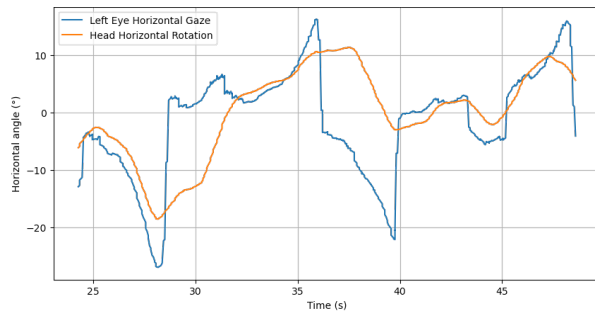


Figure 3: Example eye and head movement data recorded from a Saberception session using the Meta Quest Pro. Clear fixational VOR and saccades are visible as attention shifts between slicing one cube and looking at the next.

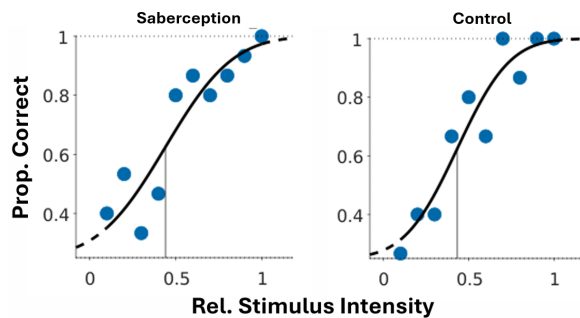


Figure 4: Exemplar psychometric functions for a single discrimination threshold measured using both the Saberception method and more traditional keyboard response control method. Vertical line indicates the perceptual threshold at 62.5% correct

ties, path properties, and the correct slice direction of each cube.

A benefit to this methodology is wealth of data which can be recorded. The platform supports the recording of eye, hand and head movements. By using predefined levels or recorded cube positions gameplay can also be replayed and information extracted from the graphics engine. The platform includes tools to output RGB images, depth maps, and normals for each frame observed by participants (see Figure 2). While there is some limitation to what experiments can be designed depending on the available VR display technology, it is important to note that modern headset can be well calibrated and can be suitable for perceptual research [4].

The Saberception platform is available open source under MIT license at <https://github.com/jmehh/Saberception>.

3 PSYCHOPHYSICAL MEASUREMENTS

To demonstrate the utility of the Saberception platform for conducting perceptual research we chose measure colour discrimination thresholds on the isoluminant Derrington–Krauskopf–Lennie (DKL) [3] colour plane. We replicated a standard 4-AFC colour patch discrimination task in which participants had to identify the position of a single differently coloured patch from three other reference patches [5]. We measure thresholds both using our Saberception platform and in a more traditional control setup, both presented on a colour calibrated Meta Quest 3 headset. Figure 1 demonstrates the Saberception task. Participants had to slice 1 cube per second. Each cube had a short ‘jump’ phase starting 1 second before it was within reach of the participant. Patches only showed for the first 500ms of this phase. The correct slice direction was

determined by the edge of the cube that contained the odd-one-out patch. For the traditional control condition, cubes appeared at mid-point of the jump (the same position as the patches would blank in the Saberception task) for 500ms. In this condition, participants recorded responses with arrow keys instead of slicing.

Using method-of-constant stimuli we were able to compute psychometric functions for colour discrimination thresholds. A single 2.30 minutes song could contain 100 trials, 10 repetitions of 10 stimulus intensity levels. Alternatively (as in Figure 1) levels could be split into left and right streams to simultaneously measure two thresholds (e.g. separate colours). Figure 4 shows exemplar psychometric functions for discrimination of chroma only shifts from a purple-ish reference colour. In pilot tests ($N=10$) we found general agreement between thresholds measured via Saberception ($M = 0.34$) and the control setup ($M = 0.33$) (paired t-test, $t = -0.2321$; $P = 0.8212$; Cohen’s $d = -0.0734$). Further, data quality was comparable between the two.

4 DISCUSSION

We successfully measured colour discrimination thresholds using our VR methodology. We were reliably able to measure psychometric curves and compute discrimination thresholds in approximately 2.5 - 4 minutes for a single direction. We expect even faster measurements using adaptive staircase methods such as QUEST. This result suggests that our Saberception methodology is suitable for rapid measurement of perceptual thresholds. While we explored colour perception to demonstrate the method, we see no reason as to why this could not be replicated for lightness perception, contrast sensitivity, or other perceptual dimensions.

While we have yet to explicitly measure engagement and enjoyment for this task, it was clear that pilot participants found the task easy and fun. We hypothesize that such engagement would benefit data collection during longer experiments where strain and attention lapses can significantly affect the quality of collected data.

We ultimately see this platform as a method for exploring perception under more active task demands. This is complimentary to other gamified and/or immersive VR methodologies [1], as well as recent methods such as continuous psychophysics [2].

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